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COURSE OUTCOME-5

DWDM METRO LINK COMMISSIONING: FROM FIELD MEASUREMENTS TO A STANDARDS-BASED UPGRADE DECISION

1. INTRODUCTION

Dense Wavelength Division Multiplexing (DWDM) is an optical communication technology that allows multiple signals to be transmitted simultaneously through a single optical fibre using different wavelengths.

DWDM technology is widely used in metropolitan communication networks, data centres, telecommunications and high-speed internet systems.

This case study examines a 60 km optical fibre link connecting two data centres. The existing system transmits eight channels at 10 Gb/s using Non-Return-to-Zero (NRZ) modulation.

During commissioning, the system experiences optical power loss, signal distortion and increased Bit Error Rate (BER).

The objective is to identify the faults, improve the optical link performance and upgrade the total transmission capacity to 1.6 Tb/s without replacing the existing optical fibre.

2. AIM

To analyze, diagnose and optimize a 60 km DWDM optical communication link using optical measurements, OTDR fault localization, eye-diagram analysis, BER calculations and ITU-T G.694.1 wavelength planning.

3. OBJECTIVES

1. To analyze optical fibre attenuation and transmission losses.
2. To calculate received optical power and system margin.
3. To identify optical fibre faults using OTDR measurements.
4. To evaluate signal quality using eye diagrams and BER.

5. To calculate dispersion and polarization mode dispersion.
6. To study quantum detection limits in optical receivers.
7. To compare different DWDM upgrade technologies.
8. To design an energy-efficient 1.6 Tb/s optical communication system.

STAGE 1 – INTERPRET THE MEASUREMENT AND PERFORMANCE REQUIREMENTS

STAGE 2 – LINK BUDGET, DISPERSION AND OSNR ANALYSIS

2.1 Fibre Attenuation Calculation

Given:

Fibre length, L = 60 km

Attenuation coefficient, $\alpha = 0.21$ dB/km

Formula:

Fibre attenuation = $\alpha \times L$

$= 0.21 \times 60$

Fibre attenuation = 12.6 dB

2.2 Design Link Loss

Given:

Fibre attenuation = 12.6 dB

Connector loss = $4 \times 0.5 = 2$ dB

Splice loss = $13 \times 0.05 = 0.65$ dB

Demultiplexer loss = 6 dB

Formula:

Total design loss = Fibre loss + Connector loss + Splice loss + Demultiplexer loss

$= 12.6 + 2 + 0.65 + 6$

Total design loss = 21.25 dB

2.3 As-Built Link Loss

Measured connector losses:

Three normal connectors = 3 × 0.5 = 1.5 dB

Faulty connector = 1.20 dB Total

connector loss = 2.70 dB

Measured splice losses:

12 fusion splices = 12 × 0.05 = 0.60 dB

Mechanical splice = 0.62 dB

Total splice loss = 1.22 dB

Therefore:

Total as-built loss = 12.6 + 2.70 + 1.22 + 6

Total as-built loss = 22.52 dB

Excess loss:

= 22.52 − 21.25

Excess loss = 1.27 dB

The measured optical link has 1.27 dB additional loss compared with the design.

2.4 Received Optical Power

Given:

Launch power = +1 dBm As-built

link loss = 22.52 dB

Formula:

Received power = Launch power − Total loss

= 1 − 22.52

Received optical power = −21.52 dBm

2.5 Chromatic Dispersion

Given:

D = 17 ps/(nm·km)

L = 60 km

Source spectral width = 0.03 nm

Bit rate = 10 Gb/s

Total pulse broadening:

$\Delta t = D \times L \times \Delta\lambda$

$= 17 \times 60 \times 0.03$

$\Delta t = 30.6$ ps

The dispersion penalty is calculated using:

$\delta = 5 \log_{10}[1 + 2\pi(BDL\sigma\lambda)^2]$ Using

the given values:

Dispersion penalty ≈ 1.00 dB

2.6 Extinction Ratio Penalty

The extinction ratio penalty can be estimated using:

$P_ER = 10 \log_{10}[(r + 1)/(r - 1)]$ where:

r = 10^(ER/10) For

Channel 3:

ER = 9 dB

Extinction ratio penalty ≈ 1.10 dB

For Channel 5:

ER = 4.5 dB

Extinction ratio penalty ≈ 3.22 dB

Channel 5 has a significantly higher extinction ratio penalty.

2.7 Effective Received Power

Formula:

Effective received power = Received power − Dispersion penalty − ER penalty

Channel 3:

$= -21.52 - 1.00 - 1.10$

Effective received power = −23.62 dBm

Channel 5:

$= -21.52 - 1.00 - 3.22$

Effective received power = −25.74 dBm

2.8 System Margin

Formula:

System margin = Effective received power − Receiver sensitivity

Channel 3:

= −23.62 − (−24)

System margin = 0.38 dB

Channel 5:

= −25.74 − (−24)

System margin = −1.74 dB

Both channels fail to achieve the required 3 dB design margin.

Channel 5 also falls below the receiver sensitivity.

The Channel 5 calculation is a diagnostic estimate because the specified receiver sensitivity is only guaranteed when the extinction ratio is at least 8.2 dB.

2.9 OSNR Calculation

Given:

EDFA input power = −6 dBm

Noise figure = 5 dB

Formula:

OSNR = 58 + P_in − NF

= 58 − 6 − 5

OSNR = 47 dB

Required OSNR = 20 dB

OSNR margin:

= 47 − 20

OSNR margin = 27 dB

Therefore, the calculated OSNR is sufficient under the simplified model specified in the case study.

The main limitation is optical power loss and signal penalties rather than the calculated amplifier noise.

The OSNR must still be verified using an optical spectrum analyzer.

2.10 Polarization Mode Dispersion

Given:

PMD coefficient = 0.1 ps/√km

Length = 60 km Formula:

PMD = 0.1 × √60

PMD = 0.775 ps At

10 Gb/s:

Bit period = 100 ps

Permitted PMD = 10% × 100

= 10 ps

Since 0.775 ps < 10 ps, the PMD is within the specified tolerance.

DESIGN DECISION 1 – LINK BUDGET CONFIGURATION

I recommend replacing the faulty mechanical splice and high-loss connector because they introduce an additional 1.27 dB loss.

After restoring the splice and connector losses to their design allowances:

Repaired link loss = 21.25 dB

Received power at +1 dBm launch = −20.25 dBm

Channel 3 effective power ≈ −22.35 dBm

Channel 3 margin ≈ 1.65 dB

Therefore, repairing the fibre alone is insufficient.

The launch power should be increased to approximately +3 dBm per channel, subject to transmitter, amplifier and nonlinear power limits.

After repair and launch-power adjustment, Channel 3 has an estimated margin of approximately 3.65 dB.

Channel 5 must also have its extinction ratio corrected before acceptance.

STAGE 3 – OTDR, SPLICE AND CONNECTOR ANALYSIS

3.1 Introduction to OTDR

An Optical Time Domain Reflectometer is used to identify faults and measure attenuation along an optical fibre.

It operates by transmitting short optical pulses into the fibre and measuring the light returning through Rayleigh backscattering and Fresnel reflection.

OTDR helps identify:

- Fibre breaks
- Connector reflections
- Splice losses
- Bending losses
- Fault locations

3.2 OTDR Event Analysis

Event	Distance	Loss	Interpretation
Event 1	12.00 km	0.05 dB	Normal fusion splice
Event 2	31.50 km	0.62 dB	High-loss mechanical splice
Event 3	59.90 km	1.20 dB	Faulty reflective connector
Event 4	59.909 km	Not resolved	Second patch-cord connector

Event 2 produces excessive splice loss.

Event 3 produces high connector loss and reflection.

Event 4 is located only 9 m from Event 3, making its loss difficult to measure with longer OTDR pulses.

3.3 OTDR Dynamic Range

Required dynamic range:

DR_required = One-way loss + 6 dB

= 22.52 + 6

Required dynamic range = 28.52 dB

Available OTDR settings:

Pulse width	Dynamic range	Event dead zone	Attenuation dead zone
10 ns	20 dB	1 m	4 m
100 ns	27 dB	5 m	15 m
1 μs	33 dB	40 m	100 m

The 1 μs pulse provides sufficient dynamic range for the complete link.

However, it cannot separate connectors located 9 m apart.

Therefore, multiple pulse widths and bidirectional testing are required.

3.4 Spatial Pulse Length

Formula:

Spatial pulse length = ct/n

Using:

c = 3×10^8 m/s n

= 1.4682

For 10 ns:

Spatial pulse length ≈ 2.04 m For

100 ns:

Spatial pulse length ≈ 20.43 m For

1 μs:

Spatial pulse length ≈ 204.33 m

These represent the physical round-trip-equivalent pulse lengths. The corresponding one-way distance resolutions are approximately 1.02 m, 10.22 m and 102.17 m.

The 10 ns pulse has sufficient dead-zone performance to resolve both connectors.

The 100 ns pulse can distinguish their reflective peaks under the specified event dead-zone limit, but cannot accurately measure the second connector's loss because its attenuation dead zone is 15 m.

The 1 μs pulse cannot resolve the two connectors separately.

3.5 Round-Trip Delay

Formula: t

= 2nL/c

For Event 1: L

= 12 km **t ≈**

117.46 μs For

Event 2: L =

31.50 km **t ≈**

308.32 μs For

Event 3: L =

59.90 km **t ≈**

586.30 μs For

Event 4: L =

59.909 km **t ≈**

586.39 μs

3.6 Location Uncertainty

Given group-index uncertainty:

Δn = ±0.002

Approximate location uncertainty:

ΔL = L(Δn/n)

For Event 2: ΔL

≈ ±42.9 m For

Event 3:

ΔL ≈ ±81.6 m

This uncertainty must be considered during physical fault localization.

3.7 Fusion Splicing Versus Mechanical Splicing

Fusion Splicing	Mechanical Splicing
Permanently joins optical fibres	Aligns fibres mechanically
Generally lower insertion loss	Generally higher insertion loss
Low reflection	Higher reflection risk
Better long-term reliability	Suitable for temporary repairs
Preferred for permanent metro links	Useful for quick restoration

The defective mechanical splice should be replaced with a fusion splice.

3.8 Connector Selection

LC connectors are suitable for high-density DWDM patch panels because of their compact size.

LC/APC connectors can provide low back-reflection when the entire mating interface is designed for APC.

LC/UPC connectors are also widely used in optical communication equipment.

UPC and APC connectors must not be directly mated because their end-face geometries are incompatible.

The connector selection must match the installed transceiver and patch-panel interface.

DESIGN DECISION 2 – OTDR FAULT LOCALIZATION

I recommend repairing Event 3 first because its 1.20 dB loss and −22 dB reflectance indicate a major connector defect.

Event 2 should receive the second repair priority because its mechanical splice introduces 0.62 dB loss.

The recommended OTDR procedure is:

1. Use a 1 μs pulse for complete-link testing.
2. Use a 10 ns pulse for detailed connector inspection.

3. Perform bidirectional OTDR measurements.
4. Use suitable launch and receive fibres.
5. Replace the defective connector.
6. Replace the mechanical splice with a fusion splice.

STAGE 4 – EYE DIAGRAM, BER AND QUANTUM LIMIT ANALYSIS

4.1 Bit Error Rate

BER represents the probability of receiving an incorrect bit during digital communication.

Formula:

$BER = \frac{1}{2} \operatorname{erfc}(Q/\sqrt{2})$

4.2 Channel 3 BER Calculation

Given:

$Q = 7.6$

$BER = \frac{1}{2} \operatorname{erfc}(7.6/\sqrt{2})$

$BER \approx 1.48 \times 10^{-14}$

This is lower than the required BER of 10^{-12}.

Therefore, Channel 3 satisfies the BER requirement.

4.3 Channel 5 BER Calculation

Given:

$Q = 4.7$

$BER = \frac{1}{2} \operatorname{erfc}(4.7/\sqrt{2})$

$BER \approx 1.30 \times 10^{-6}$

This is much higher than the required BER.

Therefore, Channel 5 fails the BER requirement.

4.4 Total Jitter Calculation

Formula:

$TJ = DJ + 14.07 \times RJ$

For Channel 3:

$DJ = 0.18$ UI

RJ = 0.012 UI

TJ = 0.18 + 14.07(0.012)

TJ = 0.349 UI

Eye width:

= 1 − 0.349

Eye width = 0.651 UI

For Channel 5:

DJ = 0.30 UI

RJ = 0.022 UI

TJ = 0.30 + 14.07(0.022)

TJ = 0.610 UI Eye

width:

= 1 − 0.610

Eye width = 0.390 UI

4.5 Comparison of Both Channels

Parameter	**Channel 3**	**Channel 5**
Q-factor	7.6	4.7
Extinction ratio	9 dB	4.5 dB
Deterministic jitter	0.18 UI	0.30 UI
Random jitter	0.012 UI	0.022 UI
Total jitter	0.349 UI	0.610 UI
Eye width	0.651 UI	0.390 UI
BER	1.48×10^{-14}	1.30×10^{-6}
Result	Passes BER target	Fails

Channel 5 experiences significantly greater signal distortion.

The low extinction ratio increases its power penalty, while higher jitter reduces the available timing margin.

Because Channel 3 performs well over the same fibre, the additional Channel 5 impairment is likely to be associated with its transmitter or channel-specific equipment.

4.6 Quantum Detection Limit

Quantum detection limits describe the minimum number of photons required to detect optical information reliably.

For ideal direct detection:

BER = ½ exp(−N) At

BER = 10^{-12}:

10^{-12} = ½ exp(−N)

N = −ln(2 × 10^{-12})

N ≈ 26.94 photons per bit

4.7 Quantum-Limited Sensitivity

Photon energy:

E = hc/λ

Using:

h = 6.626 × 10^{-34} J·s

c = 2.998 × 10^{8} m/s λ

= 1550 nm

Photon energy: **E**

≈ 1.282 × 10^{-19} J

At 10 Gb/s:

P = N × E × B P

≈ 3.45 × 10^{-8} W

Converting to dBm:

Ideal direct-detection sensitivity ≈ −44.62 dBm

Practical receiver sensitivity = −24 dBm

Difference:

$= -24 - (-44.62)$

Quantum-limit gap ≈ 20.62 dB

The practical receiver requires significantly more optical power than the ideal photoncounting receiver.

4.8 Comparison of Detection Methods

Detection method	Photons per bit	Ideal sensitivity
Direct detection	26.94	−44.62 dBm
Homodyne PSK	12.37	−48.00 dBm
Heterodyne PSK	24.74	−44.99 dBm

These values are calculated using the ideal detection formulas supplied in the assignment.

Homodyne coherent detection provides the best ideal sensitivity among the three methods.

Coherent optical communication can also improve spectral efficiency and support advanced modulation formats.

DESIGN DECISION 3 – EYE AND BER PERFORMANCE

I recommend replacing or recalibrating the Channel 5 transmitter because its extinction ratio of 4.5 dB and Q-factor of 4.7 are below the required limits.

The transmitter should be adjusted to achieve:

- Extinction ratio ≥ 8.2 dB
- Q-factor ≥ 7.03
- BER $\leq 10^{-12}$
- Improved eye opening
- Reduced deterministic and random jitter

After correction, the eye diagram and BER must be measured again.

STAGE 5 – INTEGRATED WDM UPGRADE AND ENERGY STRATEGY

5.1 Introduction

The existing DWDM system provides a total transmission capacity of 80 Gb/s.

The required upgraded capacity is 1.6 Tb/s.

Three upgrade options are considered:

Option P: 16 × 100G DP-QPSK

Option Q: 8 × 200G DP-16QAM

Option R: 4 × 400G silicon-photonic pluggables

All three options provide 1.6 Tb/s total capacity.

5.2 ITU-T G.694.1 Frequency Calculation

Given:

Reference frequency = 193.1 THz

Channel number n = +3

Grid spacing = 50 GHz = 0.05 THz

Formula:

f = 193.1 + n(0.05) f

= 193.1 + 3(0.05)

Frequency = 193.25 THz 5.3

Wavelength Calculation

Formula:

$\lambda = c/f$ $\lambda = (2.998 \times 10^{8})/(193.25$

$\times 10^{12})$

Wavelength ≈ 1551.36 nm 5.4

Flex-Grid Slot Calculation

For Option Q:

Symbol rate = 32 GBd

Roll-off factor = 0.1

Required Nyquist bandwidth:

BW = 32 × (1 + 0.1)

BW = 35.2 GHz

Flex-grid slot width = 12.5 GHz

Number of slots:

= 35.2/12.5

= 2.816

Therefore, each channel requires:

3 slots = 37.5 GHz

For eight channels:

Total occupied spectrum = 8 × 37.5

Total spectrum = 300 GHz 5.5

Spectral Efficiency

Formula:

Spectral efficiency = Total bit rate / Occupied bandwidth

Option P

Capacity = 1600 Gb/s

Bandwidth = 16 × 50 = 800 GHz

Spectral efficiency:

= 1600/800

2 bit/s/Hz Option

Q

Capacity = 1600 Gb/s

Bandwidth = 8 × 37.5 = 300 GHz

Spectral efficiency:

= 1600/300

5.33 bit/s/Hz Option

R

Capacity = 1600 Gb/s

Bandwidth = 4 × 75 = 300 GHz

Spectral efficiency:

= 1600/300

5.33 bit/s/Hz

Options Q and R provide higher spectral efficiency than Option P.

5.6 OSNR Margin Comparison

Using the calculated OSNR of 47 dB:

Option	Required OSNR	Calculated OSNR margin
P	12.5 dB	34.5 dB
Q	19.5 dB	27.5 dB
R	26 dB	21 dB

All three options have positive margins under the simplified OSNR calculation.

However, these values do not guarantee successful coherent transmission. Actual postupgrade OSNR, nonlinear penalties and receiver performance must be measured.

5.7 Power Consumption

Option P

16 modules × 22 W

Total power = 352 W

Option Q

8 modules × 20 W

Total power = 160 W

Option R

4 modules × 15 W

Total power = 60 W

5.8 Energy per Bit

Formula:

Energy per bit = Power consumption / Data rate

Option P

$= 352/(1.6 \times 10^{12})$

Energy per bit = 220 pJ/bit

Option Q

$= 160/(1.6 \times 10^{12})$

Energy per bit = 100 pJ/bit

Option R

$= 60/(1.6 \times 10^{12})$

Energy per bit = 37.5 pJ/bit

Legacy 10G Module Power

= 3.5 W

Data rate = 10 Gb/s Energy

per bit:

$= 3.5/(10 \times 10^{9})$

Energy per bit = 350 pJ/bit

These calculations compare the specified module power only, not the complete system power including amplifiers and supporting equipment.

5.9 Comparison of Upgrade Options

Parameter	Option P	Option Q	Option R
Total capacity	1.6 Tb/s	1.6 Tb/s	1.6 Tb/s

Number of channels	16	8	4
Modulation/technology	DP-QPSK	DP-16QAM	Si-photonic
Occupied spectrum	800 GHz	300 GHz	300 GHz
Spectral efficiency	2	5.33	5.33
OSNR requirement	12.5 dB	19.5 dB	26 dB
Calculated OSNR margin	34.5 dB	27.5 dB	21 dB
Module power	352 W	160 W	60 W
Energy per bit	220 pJ	100 pJ	37.5 pJ

5.10 Role of Silicon Photonics

Silicon photonics integrates optical components onto compact semiconductor chips.

Its advantages include:

1. Reduced device size.
2. Lower module power consumption.
3. High-speed optical transmission.
4. Improved component integration.
5. Reduced interconnection complexity.
6. Support for high-capacity data-centre communication.

DESIGN DECISION 4 – INTEGRATED OPTIMIZATION STRATEGY

I recommend Option R as the preferred energy-efficient upgrade candidate because it provides 1.6 Tb/s using only four 400G modules.

It requires 300 GHz of optical spectrum and consumes 60 W of specified module power.

Its energy consumption is only 37.5 pJ/bit, which is lower than Options P and Q.

However, Option R has the highest required OSNR of 26 dB. It should therefore be deployed only after the repaired link passes a full optical power, OSNR and transmission-quality verification.

Option Q is the preferred alternative if Option R cannot satisfy its actual receiver and transmission requirements.

STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE LINK

6.1 Final Verification Table

The following values are calculated predictions. Final acceptance requires field measurements after repair and transmitter adjustment.

Parameter	**Required value**	**Predicted or calculated value**	**Result**
Fibre length	60 km	60 km	Pass
Optical link loss	≤ 21.25 dB	21.25 dB after repair	Predicted pass
Launch power	Sufficient for margin	+3 dBm proposed	Requires verification
Received power	≥ −21 dBm before penalties	−18.25 dBm	Predicted pass
Channel 3 effective power	≥ −21 dBm	−20.35 dBm	Predicted pass
Channel 3 margin	≥ 3 dB	3.65 dB	Predicted pass
Channel 5 extinction ratio	≥ 8.2 dB	Requires correction	Pending
Channel 5 BER	≤ 10^{-12}	1.30×10^{-6} before repair	Fail
OSNR	Meets selected option	47 dB calculated	Field test required
OTDR events	Within loss allowances	Repair proposed	Pending

Upgrade capacity	1.6 Tb/s	1.6 Tb/s	Design pass
Upgrade energy	Minimum practical value	37.5 pJ/bit, Option R	Best of options

6.2 Verification Procedure

The following tests should be performed before deployment.

1. Measure optical fibre loss using an optical power meter.
2. Perform bidirectional OTDR testing.
3. Confirm that the repaired splice and connector losses meet the design allowances.
4. Measure received optical power at the receiver.
5. Verify optical signal-to-noise ratio using an OSA.
6. Measure Channel 5 extinction ratio and eye opening.
7. Verify Q-factor and BER using appropriate test equipment.
8. Confirm dispersion and PMD performance.
9. Install the selected DWDM upgrade modules.
10. Repeat OSNR, receiver power and BER measurements after upgrading.

FINAL SYSTEM DESIGN

7.1 Proposed System Configuration

The proposed system uses the existing 60 km G.652.D optical fibre.

The defective connector and mechanical splice are repaired before the upgrade.

The Channel 5 transmitter is corrected to improve its extinction ratio and BER.

The optical launch power is adjusted to maintain the required receiver margin.

After successful verification, the existing 80 Gb/s system is upgraded to 1.6 Tb/s using the selected DWDM technology.

8. FINAL ENGINEERING DECISION

Would You Deploy the Optimized Design?

Decision: YES, subject to successful repair and final acceptance testing.

The optimized DWDM communication system is technically suitable for upgrading the existing 60 km optical fibre link from 80 Gb/s to 1.6 Tb/s without replacing the fibre.

The initial link experiences excessive optical loss because of a defective connector and mechanical splice.

The measured link loss is 22.52 dB, which exceeds the design value of 21.25 dB.

Replacing the defective components can reduce the optical loss by approximately 1.27 dB.

Channel 5 also experiences signal degradation due to its low extinction ratio and increased jitter.

Correcting the Channel 5 transmitter is necessary to achieve the required Q-factor and BER.

After repairing the optical link and adjusting the launch power to +3 dBm, Channel 3 is predicted to achieve a system margin of approximately 3.65 dB.

For the capacity upgrade, Option R provides the lowest specified module power consumption and energy per bit.

The selected system supports:

- Total capacity of 1.6 Tb/s.
- Four 400 Gb/s optical channels.
- Occupied optical spectrum of 300 GHz.
- Spectral efficiency of 5.33 bit/s/Hz.
- Module power consumption of 60 W.
- Energy consumption of 37.5 pJ/bit.

The calculated OSNR margin is positive, but actual transmission performance must be confirmed after installation.

Therefore, deployment is recommended only after OTDR, optical power, OSNR, eyediagram and BER testing confirms that all acceptance requirements are satisfied.

9. CONCLUSION

This case study investigated the performance evaluation, fault identification, optimization, and future upgrading of a 60 km Dense Wavelength Division Multiplexing (DWDM) optical communication link. The study focused on analyzing the major factors affecting optical transmission quality, including fiber attenuation, connector losses, mechanical splice losses, transmitter degradation, receiver sensitivity, optical signal-to-noise ratio (OSNR), bit error rate (BER), and overall system reliability. The analysis demonstrated the importance of maintaining an adequate optical power budget to achieve reliable high-speed data transmission over long distances. By combining theoretical calculations with practical optical testing methods, the study provided a systematic approach for identifying transmission problems and developing suitable corrective measures. The results highlighted that even

small losses introduced by connectors, splices, and other passive optical components can significantly affect the performance of a long-distance optical communication system.

The optical link-budget analysis identified excessive connector loss, mechanical splice loss, and Channel 5 transmitter degradation as the major causes of reduced system performance. These additional losses decreased the optical power available at the receiver and prevented the existing system from achieving the required 3 dB receiver power margin. Maintaining this margin is essential for ensuring stable communication under varying operating conditions, component aging, temperature fluctuations, and possible future increases in transmission losses. The analysis emphasized the need for proper fiber installation, high-quality connectors, reliable splicing techniques, and periodic maintenance to minimize unnecessary optical losses. Corrective measures such as cleaning or replacing defective connectors, repairing high-loss splices, and restoring transmitter output power were identified as suitable methods for improving the optical power budget. These improvements can increase the available receiver margin, reduce the possibility of signal degradation, and enhance the long-term reliability of the DWDM network.

The Optical Time Domain Reflectometer (OTDR) analysis played an important role in identifying the locations of defective optical components along the 60 km fiber link. By examining reflected and backscattered optical signals, OTDR measurements helped distinguish between normal fiber attenuation and abnormal losses caused by connectors, splices, and other discontinuities. This approach provided a practical method for locating faults without requiring complete dismantling of the optical transmission system. The identified fault locations supported the development of a targeted repair strategy, reducing unnecessary maintenance activities and improving troubleshooting efficiency. The study also demonstrated that regular OTDR testing can support preventive maintenance by identifying gradual deterioration in fiber connections before it causes major communication failures. Proper documentation of OTDR measurements and comparison with baseline results can further improve network monitoring and fault management.

The eye-diagram and Bit Error Rate (BER) analysis provided additional information about the quality and reliability of the transmitted optical signals. The results indicated that Channel 3 satisfies the specified BER target, demonstrating acceptable signal quality under the evaluated conditions. However, Channel 5 requires transmitter correction because its degraded performance affects the reliability of data transmission. Eye-diagram evaluation is useful for understanding signal distortion, timing variations, noise effects, and the available decision margin at the receiver. A clear eye opening generally indicates better signal quality, while a reduced eye opening suggests increased distortion or noise. The study highlighted the importance of combining BER measurements with eye-diagram analysis to evaluate individual DWDM channels and identify performance differences. Correcting the Channel 5 transmitter and verifying its output power and signal quality are necessary steps before confirming that the channel meets the required communication performance.

The quantum-limit calculations provided a theoretical reference for evaluating optical receiver sensitivity and understanding the fundamental limitations of optical detection. The comparison between ideal quantum-limited performance and practical receiver sensitivity demonstrated that real optical receivers are affected by additional factors such as thermal

noise, shot noise, detector efficiency, electronic circuit limitations, and implementation losses. These factors increase the optical power required to achieve a specified BER compared with an ideal receiver. The analysis emphasized that receiver sensitivity depends not only on the optical power received but also on detector characteristics, noise performance, modulation format, and signal-processing techniques. Understanding these limitations is important for selecting suitable optical receivers and designing communication systems that provide reliable performance under practical operating conditions.

The study also evaluated three DWDM upgrade options by comparing their transmission capacity, occupied optical spectrum, OSNR requirements, and energy consumption. These parameters are essential for selecting an appropriate network upgrade strategy because modern optical communication systems must support increasing data traffic while maintaining high reliability and energy efficiency. Higher-capacity transmission technologies can improve spectral utilization, but they may also require more complex modulation techniques, improved receiver performance, and higher OSNR. Therefore, selecting an upgrade option requires a balanced evaluation of transmission performance, equipment complexity, power consumption, and future scalability. The comparison demonstrated that the highest transmission capacity is not necessarily the most suitable solution when energy efficiency and practical deployment requirements are considered.

Based on the specified comparison criteria, Option R was selected as the preferred energy-efficient DWDM upgrade solution because it provides a total transmission capacity of 1.6 Tb/s with the lowest specified module power consumption among the evaluated alternatives. This option offers a suitable balance between high-speed data transmission and reduced equipment energy requirements. Lower module power consumption can contribute to reduced operating costs, lower heat generation, and improved sustainability in optical communication infrastructure. The selected option also provides a foundation for supporting growing network traffic and future communication demands. However, successful deployment requires verification of its OSNR requirements, compatibility with the existing optical infrastructure, available spectral resources, and practical operating conditions. Proper installation, testing, and commissioning are necessary to ensure that the upgraded system achieves its expected performance.

Furthermore, the case study emphasized the importance of adopting a systematic approach to optical network maintenance and performance optimization. Regular monitoring of optical power levels, transmitter output, receiver sensitivity, BER, OSNR, and fiber attenuation can help identify potential problems before they result in service interruptions. Preventive maintenance practices, including connector inspection, fiber cleaning, splice verification, and transmitter performance testing, can significantly improve network availability. The use of optical measurement instruments and analytical techniques also helps reduce troubleshooting time and supports accurate engineering decisions. By integrating fault diagnosis, performance evaluation, and upgrade planning, network operators can improve transmission reliability while controlling maintenance and operational costs.

The proposed repair and optimization strategy also supports the long-term scalability and sustainability of optical communication networks. As demand for high-speed internet, cloud computing, data centers, artificial intelligence applications, and advanced

telecommunications continues to increase, DWDM technology remains an important solution for expanding transmission capacity over existing optical fiber infrastructure. Efficient spectrum utilization, low-loss optical components, reliable transmitter operation, and optimized receiver performance are essential for meeting these growing requirements. Future improvements may include automated optical performance monitoring, intelligent fault detection, advanced digital signal processing, improved optical amplification, and energy-efficient transceiver technologies. These developments can further enhance system capacity, reduce transmission errors, and improve overall network efficiency.

Overall, this case study demonstrated that reliable DWDM optical communication requires the combined application of accurate link-budget calculations, OTDR fault analysis, eye-diagram evaluation, BER testing, quantum-limit analysis, and systematic upgrade planning. The investigation identified the major causes of performance degradation in the existing 60 km optical link and proposed practical corrective measures to restore acceptable operating conditions. The selection of Option R as the preferred 1.6 Tb/s energy-efficient upgrade highlights the importance of balancing transmission capacity, signal quality, power consumption, and deployment feasibility. The study concludes that proper optical measurements, preventive maintenance, effective fault correction, and carefully selected network upgrades can significantly improve the reliability, efficiency, scalability, and long-term performance of modern DWDM optical communication systems

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10. REFERENCES

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